Shot Timer Mk1 Manual

Huckleberry Innovations · Firmware v1.7.5 · Oct 18, 2026 · @Is This Ai Free

This manual covers everyday use of the Shot Timer Mk1 in Part 1 and the technical details for servicing, building and updating the timer in Part 2. If you only want to shoot, Part 1 is all you need.

Part 1: Using the timer

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1. Meet your Shot Timer

The Shot Timer Mk1 hears your shots and times them: press START, wait for the beep, shoot, then press STOP to see your times. It records up to 60 shots per string, shows the battery level in the corner of every result page, and remembers your last string even after you switch it off.

Part	What it does
Screen	Two-color display. Page titles and the battery gauge show in the yellow strip at the top; your times and readings show in the blue area below.
START button	Starts a string. Held down, it opens test mode, or on the BATTERY page, offers to mark the battery full.
STOP button	Ends a string and flips through your results. Held down, it opens the sound sensitivity setting.
Microphone	Clip-on lavalier mic that plugs into the timer's jack. It listens for shots.
Buzzer	Sounds the start beep and battery warnings.
White light	On while the timer waits for the beep, blinks while a string is running, and stays on in test mode and the sensitivity setting.
Power switch	Turns the timer on and off. With the switch off, the timer draws no power.
Battery	Rechargeable 9 V lithium pack with its own USB-C charging port. Use only this type of battery.

Throughout this manual, a **tap** means press and let go right away. A **hold** means keep the button pressed for the time given, usually 2 or 3 seconds.

2. Quick start

1. **Charge the battery** until its charge light turns green (about 1 to 2 hours).
2. **Switch the timer on.** The start screen shows **Shot Timer Mk1**, **v1.7.5** and the battery gauge.
3. **Mark the battery full:** tap STOP until you reach the **BATTERY** page, hold START for 3 seconds until **MARK FULL?** appears, then tap STOP. Tap STOP again to move on.
4. **Clip on the mic** and make sure it is plugged in.
5. **Tap START**, wait for the beep, shoot, then **tap STOP** to see your times.

3. Screen guide

These are the screens exactly as the timer draws them. The battery gauge in the top-right corner (icon plus percentage) appears on the start screen and every result page.

Start and shooting



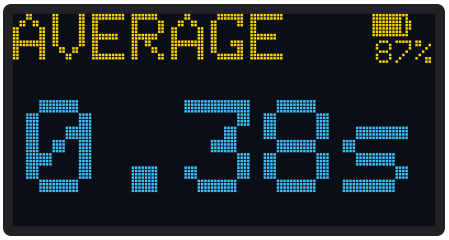
Start screen. Version on the second line, battery gauge top right.



GO! Shown with the start beep; the clock is running.



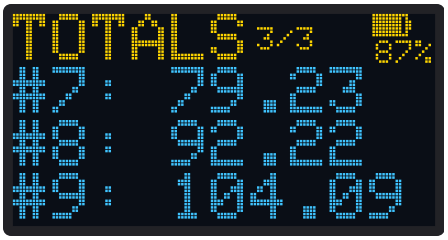
1ST SHOT. Draw time. **LAST STRING** marks a string recalled from memory.



AVERAGE. Average split after the first shot.

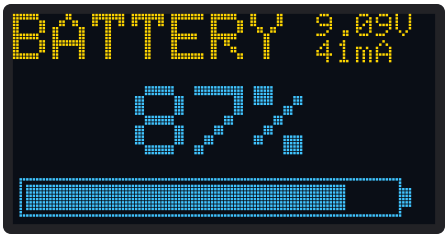


SPLITS. Time between shots, 3 per page; page number beside the title.

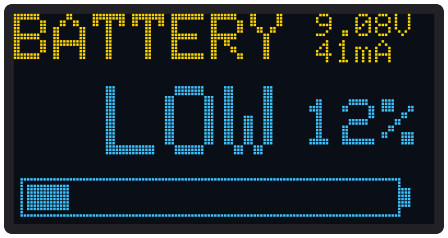


TOTALS. Time from the beep. Long strings are fine: shot 9 here at 104.09 s.

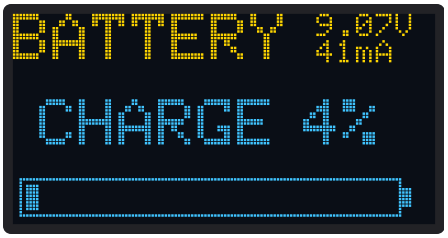
Battery



BATTERY, normal. Percentage and fill bar; live volts and current top right.



LOW (15% or less). LOW flashes beside the percentage.



CHARGE (5% or less). CHARGE flashes beside the percentage; recharge soon.



Not marked yet. Charge, then mark the battery FULL.

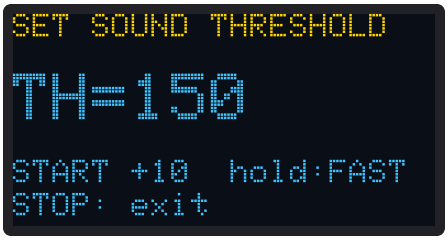


MARK FULL? After holding START 3 s. STOP = yes, START = no.

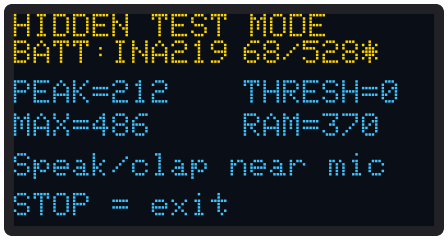


FULL. Confirmation; the gauge is reset to 100%.

Setup, service and special cases



Sound sensitivity. Hold STOP 2 s to open.



Test mode. Live mic level, battery sensor and memory readings.



Factory reset. Hold START and STOP together 3 s.



NO BATTERY. Battery switched off or running from USB only.



Long pause. A gap over 65.5 s shows as 65.53+ (at least 65.53 s).

4. Battery and charging

> **Use only the rechargeable 9 V lithium pack.** Alkaline and other disposable 9 V batteries lose voltage steadily as they are used, so the timer would show false LOW and CHARGE warnings, and beep, long before the battery is actually empty.

Charging

1. Switch the timer off and plug the battery's USB-C port into any USB charger. The battery can stay inside the timer while it charges.
2. Wait for the charge light to turn from blue to green, about 1 to 2 hours. A bigger charger does not make it faster.
3. If a USB-C to USB-C cable does not light the charge light, use the included USB-A to USB-C cable.
4. Unplug, switch the timer on, and **mark the battery FULL** (below).

Mark the battery FULL after every charge

1. Tap STOP until you reach the BATTERY page.
2. Hold START for 3 seconds. The screen asks **MARK FULL?** Let go of START; the question stays up.
3. Tap **STOP** to confirm. The screen shows **FULL** and the gauge resets to 100%. Tap **START**, or wait 5 seconds, to cancel without changing anything.

Until you do this the first time, the gauge shows **--%** and the BATTERY page shows **Hold START if full**.

Only mark FULL right after a full charge. Marking it partway through makes the gauge read higher than the real charge, so the warnings come late. If it happens by mistake, charge the pack fully and mark FULL again.

How long it lasts

A full charge runs the timer for **about 16 hours** while you use the result pages, or about 13 hours if you leave it on the BATTERY page, whose lit fill bar uses a little more power. LOW warns you about 2 to 2½ hours before the battery runs out, and CHARGE about 45 to 50 minutes before. Both were confirmed in full run-down tests.

5. Running a string

A string starts when you let go of START, beeps after a random 3 to 8 second wait, and ends when you press STOP.

1. Clip the microphone on and make sure it is plugged in.
2. **Tap START.** The countdown starts when you release the button, and the white light comes on. Keep the tap short: holding START for 2 seconds opens test mode instead.
3. **Wait for the beep.** The delay is random, between 3 and 8 seconds, so you cannot anticipate it.
4. **Shoot on the beep.** The buzzer sounds for half a second, the screen shows **GO!** and the clock starts.
5. **Shoot your string.** Each shot appears on the SPLITS page as it is heard, and the white light blinks every half second while the clock runs.
6. **Press STOP** when you are done. The clock stops the instant you press, the string is saved, and the screen jumps to your results.

Good to know:

- Up to 60 shots per string are recorded, and a string can run as long as you like. Each gap between shots (and from the beep to the first shot) is timed exactly up to 65.53 seconds; a longer pause shows as **65.53+**, meaning at least that long, and any TOTALS or AVERAGE that include it also show a **+**.
- Sounds less than 0.10 seconds after a shot are ignored, so echoes and ringing steel are not counted as extra shots.
- Once you tap START, the string cannot be cancelled during the wait, and STOP does nothing until the beep. Let it beep, then press STOP.
- Each new string replaces the one saved before it.

6. Reviewing your results

After a string, tap STOP to step through the result pages in this order; after BATTERY it wraps back to 1ST SHOT. The page changes when you **let go** of STOP, so holding STOP to open the sensitivity setting never flips a page by accident. Tap START from any page to run a new string.

Page	What it shows
1ST SHOT	Time from the start beep to your first shot, in large numbers. Shows NO SHOTS if nothing was heard.
AVERAGE	Average time between shots, not counting the draw to the first shot. Shows -- with fewer than 2 shots.
SPLITS	Time between each shot and the one before it, 3 shots per page. The page number (for example 1/3) sits beside the title.
TOTALS	Time of each shot counted from the beep, 3 shots per page.
BATTERY	Charge left, live volts and current, and warnings (see section 7).

All times are in seconds with two decimals, for example 1.42, and long strings show three-digit seconds such as 104.09. The battery gauge in the top-right corner of every page keeps you posted without leaving your results.

Recalling your last string

Your last string is kept when you switch the timer off. At power-on the start screen shows **STOP: last results**; tap STOP to open them. The 1ST SHOT page then shows **LAST STRING** at the bottom so you know it comes from memory.

7. Battery gauge and warnings

The **corner gauge** on the start screen and result pages shows a small battery icon and the percentage left; it updates as the percentage changes. The **BATTERY page** shows the same percentage in large numbers with a fill bar, plus the live battery voltage and current draw. It updates every second, so you can leave it up and watch the percentage count down (about 1% every 8 minutes on that page).

Watch the percentage, not the voltage. The battery holds a steady voltage of about 9.1 V right up until it runs out, so voltage alone gives no warning. The gauge counts the charge you use and warns you in time.

Switching off is fine. The gauge saves its count as it goes and at the end of every string, so turning the timer off between strings or stages keeps the percentage accurate to within a fraction of a percent.

The gauge learns your battery. After one full charge and run-down, its estimate matches your battery's real capacity, and it keeps adjusting as the battery ages. If the timer runs longer than the gauge expects, it holds at 1% and shows CHARGE until you charge and mark FULL again, so it never claims more charge than you have.

Warning	When it appears	On the BATTERY page	On other pages
None	Above 15% (and above 8.5 V)	Percentage and fill bar	No beeps
LOW	15% or less, or below 8.5 V	Flashing LOW beside the %, single beep every 10 s	Single beep every 60 s
CHARGE	5% or less, or below 7.9 V	Flashing CHARGE, double beep every 5 s	Double beep every 30 s
NO BATTERY	Battery switched off, or running from USB only	NO BATTERY	No beeps

Whichever of percentage or voltage is worse sets the warning. Warning beeps never sound during the start wait, during a string, in test mode, in the sensitivity setting or on the MARK FULL? screen. On pages other than BATTERY they also pause for **3 minutes after any button press**, after a string, and after power-on, so reviewing your results stays quiet; the corner gauge still shows the percentage. When you see CHARGE, finish up and recharge soon.

Early units without the built-in battery sensor show only OK, LOW or CHARGE on the BATTERY page, with no percentage or corner gauge, and beep only on that page. Test mode tells you which kind you have.

8. Adjusting sound sensitivity

The threshold sets how loud a sound must be to count as a shot: a higher number is less sensitive, a lower number more sensitive. It starts at 150, which suits most ranges.

1. From the start screen or any result page, hold STOP for 2 seconds. The screen shows **SET SOUND THRESHOLD** and the current value (**TH=**).
2. Tap START to raise the value by 10. Hold START to step quickly.
3. Past 500 the value wraps back to 0, so keep going to come around to a lower number.
4. Tap STOP once to save and exit. The setting is kept when you switch off.

- **Raise it** if the timer counts extra shots from other bays, echoes or loud talking.
- **Lower it** if it misses shots, for example with a suppressed or rimfire gun, or with the mic far away.

To pick a value, use test mode to see how loud your shots and the background noise read, then set the threshold between the two.

9. Test mode

Test mode shows what the microphone hears, live, so you can check the mic and choose a sensitivity setting.

1. From the start screen or any result page **except BATTERY**, hold START for 2 seconds. The screen shows **HIDDEN TEST MODE**.
2. Clap, talk or dry-fire near the mic and watch the readings.
3. Tap STOP once to exit.

Reading	Meaning
PEAK	How loud the sound is right now.
MAX	The loudest sound since you entered test mode.
BATT	INA219: the unit has the battery sensor and % gauge. DIVIDER: an early unit with OK/LOW/CHARGE only.
mAh figures	Charge used and the gauge's capacity, for example **68/528***. The * means the capacity has been learned from your battery.
THRESH	Always 0 here so every sound shows up; not your saved setting.
RAM	A service reading, about 370 on a healthy unit; see Part 2.

The white light stays on in test mode, so don't leave the timer there for long. If PEAK never moves when you clap, check that the mic is plugged in all the way.

10. Factory reset

A factory reset clears your saved string and sets the sound threshold back to 150.

1. From the start screen or any result page, hold START and STOP together for 3 seconds.
2. The screen shows **RESET**, **Saved data cleared** and **Threshold = 150**, then returns to the start screen.

The battery gauge's learned capacity is kept, so you do not need to run the battery down again.

11. Care and storage

- Switch the timer off when you are done. With the switch off it draws nothing.
- The battery loses about 5 to 10% of its charge a month on the shelf. Top it up every few months in storage, and don't store it fully empty.
- Charge and mark FULL before a range day so the gauge starts from a known point.
- Keep the mic cable free of sharp bends at the plug, and unplug it by the plug, not the cable.
- The case is 3D-printed plastic: keep it out of hot cars and direct summer sun for long periods.

12. Troubleshooting

Problem	What to try
Screen stays dark at power-on	Check the power switch and charge the battery. If it is still dark with a charged battery, contact Huckleberry Innovations.
Extra shots counted	Raise the sound threshold. Other shooters, echoes and steel ringing can trigger it at low settings.
Shots missed	Lower the sound threshold and check the mic is plugged in and clipped close to you. Use test mode to check the mic responds.
Shots fired very fast count as one	Shots less than 0.10 s apart are treated as one sound.
String stopped recording	The timer records up to 60 shots per string.
A time shows 65.53+	More than 65.5 seconds passed between those two shots. The + means "at least"; TOTALS and AVERAGE that include that pause show + too. All other splits are exact.
Pressing START opened test mode	You held START for 2 seconds. Tap STOP to exit, then tap START briefly.
Pressing STOP opened the sensitivity screen	You held STOP for 2 seconds. Tap STOP once to save and leave; your value is unchanged unless you tapped START.
Started a string by mistake	It cannot be cancelled during the wait. Let it beep, then tap STOP.
MARK FULL? appeared by accident	Tap START or wait 5 seconds. Nothing changes.
Gauge shows --%	The battery has not been marked full yet. Charge it, then mark FULL on the BATTERY page.
Marked FULL by mistake	The gauge now reads too high. Charge fully, then mark FULL again.
Percentage seems wrong	Charge fully and mark FULL. After one full run-down the gauge learns your battery's real capacity.
Stuck at 1% with CHARGE	The battery lasted longer than the gauge expected. Charge it and mark FULL to correct the gauge.
Voltage reads about 9 V but the percentage is low	Normal. The battery holds a steady voltage until it is nearly empty; trust the percentage.
LOW or CHARGE warnings soon after fitting a new battery	A disposable (alkaline) 9 V battery is fitted. Use the rechargeable lithium pack.
BATTERY page shows NO BATTERY	The battery is switched off or flat, or the timer is running from USB only.
Low-battery beeps stopped while I was using it	Normal. Off-page beeps pause for 3 minutes after any button press; the corner gauge still shows the level.

Charge light does not come on	Use the included USB-A to USB-C cable instead of a USB-C to USB-C cable.
Chirp from the buzzer when plugging into USB	Normal and harmless: a chirp at power-up, usually two on a computer. Handy as a sign the USB connection is live, though not every unit chirps.

For help, visit huckleberryinnovations.com. Firmware, schematic and manuals are open source at github.com/jfkure/shot-timer-mk1.

13. Quick reference

Button press	Where	What happens
Tap START	Start screen or any result page	Starts a string
Press STOP	During a string	Ends the string instantly and shows results
Tap STOP	Start screen or result pages	Opens results / next page (on release)
Hold START 2 s	Any page except BATTERY	Test mode (tap STOP to exit)
Hold START 3 s, then tap STOP	BATTERY page, right after a full charge	MARK FULL? → battery marked FULL
Hold STOP 2 s	Start screen or result pages	Sound sensitivity setting (tap STOP to save)
Hold START + STOP 3 s	Start screen or result pages	Factory reset

14. Specifications

Specification	Value
Firmware	v1.7.5, open source: github.com/jfkure/shot-timer-mk1
Start delay	Random, 3 to 8 s
Start beep	0.5 s
Shots per string	Up to 60
String length	No overall limit. Each gap timed exactly up to 65.53 s; longer gaps show as 65.53+. Longest string with every time exact: about 65 minutes (60 shots).
Minimum time between shots	0.10 s
Time display	Seconds, 2 decimals (up to 3-digit seconds)
Sound threshold	0 to 500 in steps of 10, default 150
Battery	Rechargeable 9 V lithium pack only (labeled 1300 mAh at 3.7 V), USB-C charging, about 1 to 2 h
Battery capacity	About 528 mAh at 9 V (measured over two full run-downs: 526 and 529 mAh)
Current draw	About 33 mA on result pages, 41 mA on the BATTERY page, 44 mA in test mode
Runtime	About 16 h per charge (about 13 h if left on the BATTERY page)
Low warning	15% or below 8.5 V, about 2 to 2½ h before the battery runs out
Charge warning	5% or below 7.9 V, about 45 to 50 min before the battery runs out
Shelf loss	About 5 to 10% per month; no drain from the timer with the switch off

Part 2: Technical reference

For servicing, building and updating the Shot Timer Mk1. Not needed for everyday use.

15. How it works

Power-up

1. The OLED starts, and the firmware probes the I2C bus for the INA219 battery sensor. If it answers, it becomes the battery source; otherwise the timer falls back to the A1 voltage divider (early units).
2. The sound threshold, the last string's splits and the battery gauge state are loaded from EEPROM.
3. The start screen appears with the corner gauge, plus **STOP: last results** if a saved string exists.

Shot detection and timing

- The mic input (A0) is sampled at about 4 kHz in 5 ms windows of 20 samples; each window's peak-to-peak value is compared with the threshold.
- A shot is counted on a rising edge (quiet to loud) and only if at least 100 ms have passed since the last shot, which rejects echoes and ringing steel.
- Shot times are stored as 16-bit millisecond stamps (120 bytes for 60 shots). Each gap is measured with the 32-bit clock when the shot is heard and stored as the previous stamp plus the gap, capped at 65,530 ms, so splits never wrap; a capped gap is shown as 65.53+. TOTALS and AVERAGE are built from a 32-bit running sum of the splits (no overall length limit) and carry a + if they include a capped gap. Saved strings keep the cap, so recalled strings show the + too.
- The random start delay is seeded from the moment START is released, so it does not repeat between power cycles.

Buttons

- STOP ends a running string on press. In idle, STOP taps are acted on at release and only if shorter than 2 s, so the 2 s sensitivity hold and 3 s two-button reset never flip a page first. STOP is ignored during the start wait.
- Test mode, the sensitivity menu and MARK FULL? watch STOP continuously, act on the first solid press (15 ms check), then wait for release so one tap can't trigger twice.

Design choices

- Integer-only math: the floating-point library is never linked.
- All screen text lives in flash memory (`F()`), keeping it out of RAM.
- Saved splits are read from and written to EEPROM directly, with no RAM copy.
- The INA219 is driven by direct register access rather than a library.
- USB serial logging is compiled out by default (`DEBUG_SERIAL 0`), freeing 157 bytes of RAM; see section 20.
- Every timer (blinks, beeps, refreshes, battery sampling) is non-blocking, so the buttons and mic are always responsive.

- I2C runs at 400 kHz so screen updates don't stall shot detection.

16. Hardware and wiring

Function	Pin / bus
Controller	Arduino Nano (ATmega328P), on protoboard
Microphone	A0, via the TRRS jack with a bias network
Legacy battery divider	A1 (10k / 6.8k), fallback only
OLED 128×64, SSD1306	I2C address 0x3C (A4 SDA, A5 SCL)
INA219 sensor	I2C address 0x40 (A4, A5), A0/A1 address pads open
Buzzer (PWM)	D3
STOP / START buttons	D8 / D9, input pull-up, to ground
White LED	D13 through 1k
Free pins	D2, D4-D7, D10-D12, A2, A3, A6, A7. The old 7-segment code was removed in v1.7.3, so D4-D7 and D10-D12 are no longer driven.

INA219 power path

The INA219 sits high-side in the battery line: **Battery + → power switch → divider top and INA219 Vin+**, then **INA219 Vin- → Nano VIN**. INA219 Vcc goes to the Nano 5 V, GND to common ground, SDA/SCL in parallel with the OLED. The 0.1 Ω shunt (R100) drops only a few millivolts at the timer's current.

- **Vcc goes to 5 V, never 9 V.** Only Vin+/Vin- see battery voltage.
- **Polarity:** a negative mA reading means Vin+ and Vin- are swapped.
- **No bypass:** no direct wire may remain between the switch junction and VIN, or the shunt is bypassed and current reads about 0 mA.
- **USB back-feed:** on USB power with the battery switched off, about 4 V leaks back through the Nano regulator onto VIN, where the INA219 measures. Readings below 6.0 V are therefore treated as no battery.
- The 6-pin header's Vin+/Vin- are the same nets as the screw terminal; use either.

17. Battery gauge internals

- **Coulomb counting:** current is read every 0.5 s on every screen, including during strings, and integrated into charge used. Each read takes about 0.3 ms of I2C time and does not affect shot detection.
- **Saving:** the charge-used total is written to EEPROM every 1 mAh and at the end of every string, rotated across 16 slots for wear leveling (about 33 writes per slot per full discharge, roughly 3,000 full charges of EEPROM life). Switching off loses at most 1 mAh of count; at power-up 0.5 mAh, the

average loss, is added back, so repeated power cycles stay within about ±0.5 mAh and don't drift high. The added 0.5 mAh is only stored once real use reaches the next save point, so switching on and off without use doesn't accumulate it.

- **Percentage:** 100% minus charge used as a share of the capacity estimate. It holds at 1% if use exceeds the estimate.
- **Capacity learning** (starts at 500 mAh; the first measurement replaces it, later ones blend 3:1 toward the newest):
  - Battery that sags: if a battery marked FULL stays below 7.9 V for 5 s, the charge used so far becomes the capacity.
  - Battery that holds 9 V to the end (the supplied pack): at the next FULL mark, if the discharge that just ended used at least 70% of the estimate, its count becomes the capacity. Short top-ups under 70% are ignored.
- **Corner gauge:** checked every 10 s while idle and redrawn only when the percentage changes (about every 9 to 10 minutes); never redrawn during the start wait or a string.
- **Alarms:** the off-page alarm uses the voltage cached by the gauge (no extra I2C traffic), is suppressed below 6.0 V (no battery), and pauses for 3 minutes after any button press, string end or power-on.
- **Measured battery:** the supplied pack delivered 526 mAh at about 9.1 V (about 4.8 Wh, matching its 1300 mAh at 3.7 V label) and held 9.1 V until shutoff; the gauge learned 527 mAh. A second full drain on v1.7.4 (Oct 18, 2026) measured 529 mAh, blending the learned capacity to 528 mAh. LOW and CHARGE fired on schedule and the battery shut off as the count reached capacity, confirming 0% means empty.

18. Memory map

EEPROM bytes	Contents
0 - 7	Signature, sound threshold, saved shot count
8 - 127	Last string: up to 60 splits (ms)
128 - 135	Gauge signature, learned capacity (mAh), gauge flags
136 - 263	16-slot wear-leveled ring of charge-used records

A factory reset clears bytes 0 - 127 only. EEPROM contents survive reflashing between v1.6.4 and v1.7.x.

19. Assembly and service

- **Enclosure fit:** a pinched SDA wire stops the OLED at power-up (dark screen). Route I2C wires along the walls with slack, insulate the INA219 (Kapton or heat-shrink) and fix it with foam tape so it can't short against the protoboard.
- **USB chirp:** each reset leaves D3 floating through the bootloader, so the buzzer can chirp at power-up (two chirps on a computer: power-up plus the port-open reset). This is kept on purpose as a handy

USB-connected cue; it is not guaranteed on every unit.

- **Bring-up check:** test mode shows **BATT:INA219**; the BATTERY page shows about 9 V and a positive current (about 41 mA on that page). With a fresh build, charge, mark FULL and run one full drain so the gauge learns the battery.
- **Free RAM:** test mode's RAM reading should be about 370 at idle. Below about 150, the stack can run into the end of the screen buffer while recording, which shows up as stray pixels in the bottom-right corner.

20. Programming and firmware

- **Source:** the latest firmware, schematic, enclosure files and manuals are at github.com/jfkure/shot-timer-mk1.
- Use a data-capable USB cable to a computer; a charger port or charge-only cable gives "no ports discovered".
- Close the Serial Monitor before uploading. Clone Nanos may need **Processor: ATmega328P (Old Bootloader)**.
- Libraries: Adafruit SSD1306, Adafruit GFX and Wire/EEPROM (built in). No INA219 library is needed.
- **Debug logging:** set `#define DEBUG_SERIAL 1` near the top of the sketch to print the split table and status lines at 115200 baud. Bench use only: it costs 157 bytes of RAM (free RAM drops to about 210).
- Resource use (v1.7.5, as built by the Arduino IDE): about 85% of flash; about 370 bytes of free RAM at idle.
- The same firmware runs on units with and without the INA219. Updating keeps the saved threshold, last string and learned battery capacity.

21. Firmware history

Version	Changes
v1.7.5	Gaps over 65.5 s shown as 65.53+ (with + on TOTALS and AVERAGE that include them) instead of wrapping to a wrong value; CHARGE screen shows the percentage beside the flashing CHARGE, like the LOW screen.
v1.7.4	Gauge count saved every 1 mAh (was 2) and at the end of every string; 0.5 mAh added back at power-up so repeated on/off cycles no longer make the gauge read high. Fixes: MARK FULL? no longer cancels itself from contact bounce when START is released (START must stay released 150 ms, the 5 s timeout starts on release); leaving test mode no longer starts a string.
v1.7.3	TOTALS and AVERAGE correct for strings of any length; battery gauge in the corner of the start screen and result pages (AVG SPLIT renamed AVERAGE, LAST STRING tag at the bottom); MARK FULL? confirmation; off-page battery beeps pause 3 min after activity; one-tap exit from test mode and the sensitivity menu; STOP flips pages on release; NO BATTERY on USB power; serial logging off by default (+157 bytes RAM, fixes stray pixels while recording); 7-segment code removed.
v1.7.2	Gauge learns batteries that hold 9 V until shutoff; LOW at 15% and CHARGE at 5%; warnings on every idle page.
v1.7.1	Coulomb-counting fuel gauge with percentage, FULL mark and self-calibration.
v1.7.0	INA219 battery sensing with live volts and mA; automatic fallback to the A1 divider.
v1.6.4	Single LOW beep every 10 s on the BATTERY page.
v1.6.3	Two-stage battery warning (LOW 8.5 V, CHARGE 7.9 V) with cliff buzzer.
v1.6.1 - 1.6.2	BATTERY page live refresh; 8-sample averaging of the battery reading.
v1.6.0	Last string and threshold saved to EEPROM; factory reset; removed the RAM copy of saved splits that overflowed the stack and rebooted the timer at about 11 shots.
v1.5.x	1ST SHOT and AVG SPLIT summary screens; two-color screen layout.

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